

Data Communications and Networking Fourth Edition

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Chapter 16

Wireless WANs: Cellular Telephone and Satellite Networks

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16-1 CELLULAR TELEPHONY

Cellular telephony is designed to provide communications between two moving units, called mobile stations (MSs), or between one mobile unit and one stationary unit, often called a land unit.

Topics discussed in this section:

Frequency-Reuse Principle
Transmitting
Receiving
Roaming
First Generation
Second Generation
Third Generation

Figure 16.1 *Cellular system*

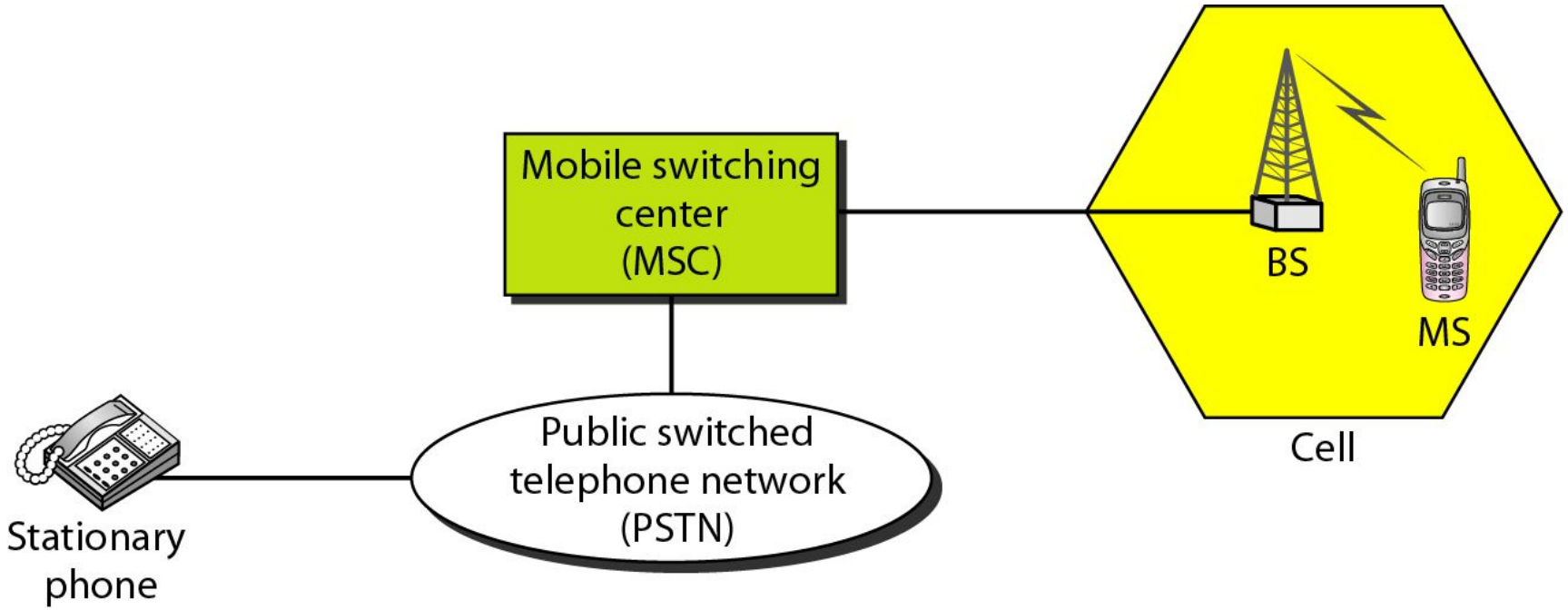
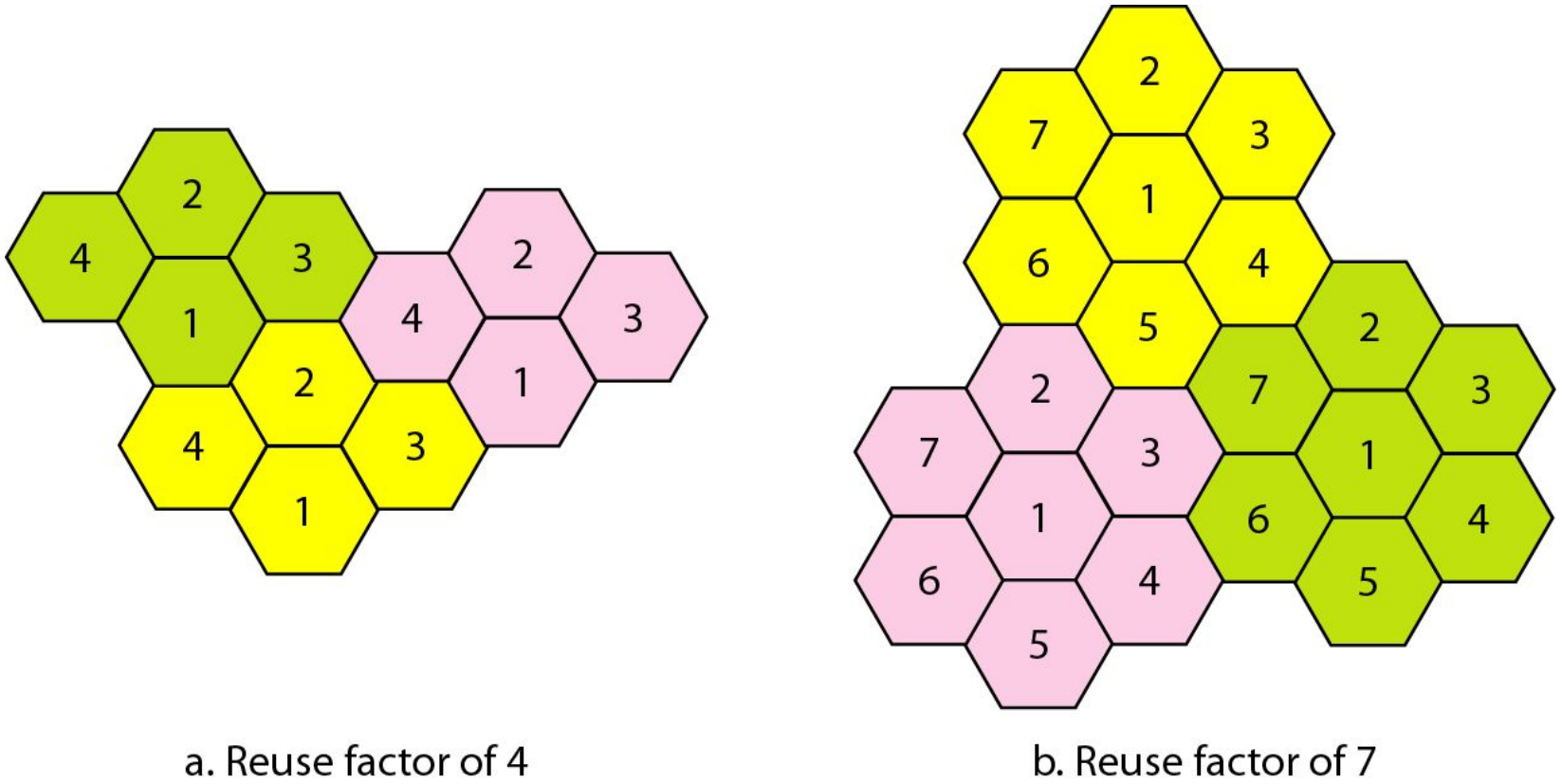


Figure 16.2 *Frequency reuse patterns*



a. Reuse factor of 4

b. Reuse factor of 7



Note

AMPS is an analog cellular phone system using FDMA.

Figure 16.3 *Cellular bands for AMPS*

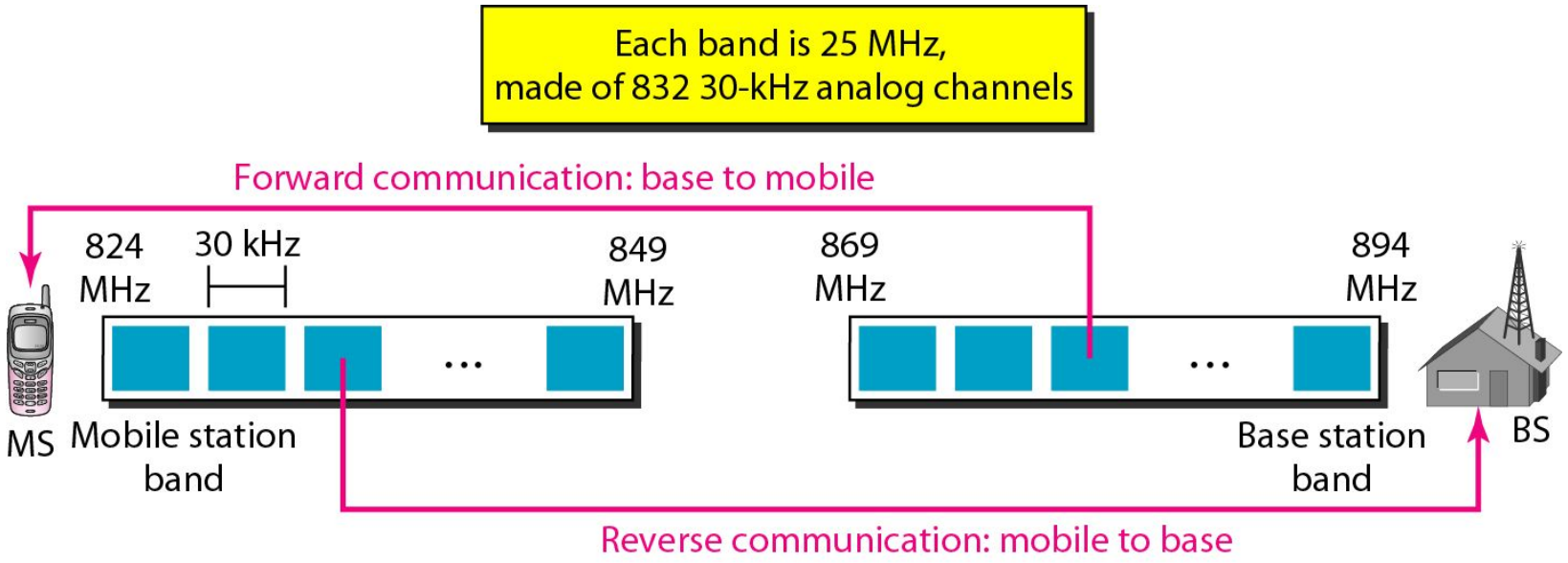


Figure 16.4 *AMPS reverse communication band*

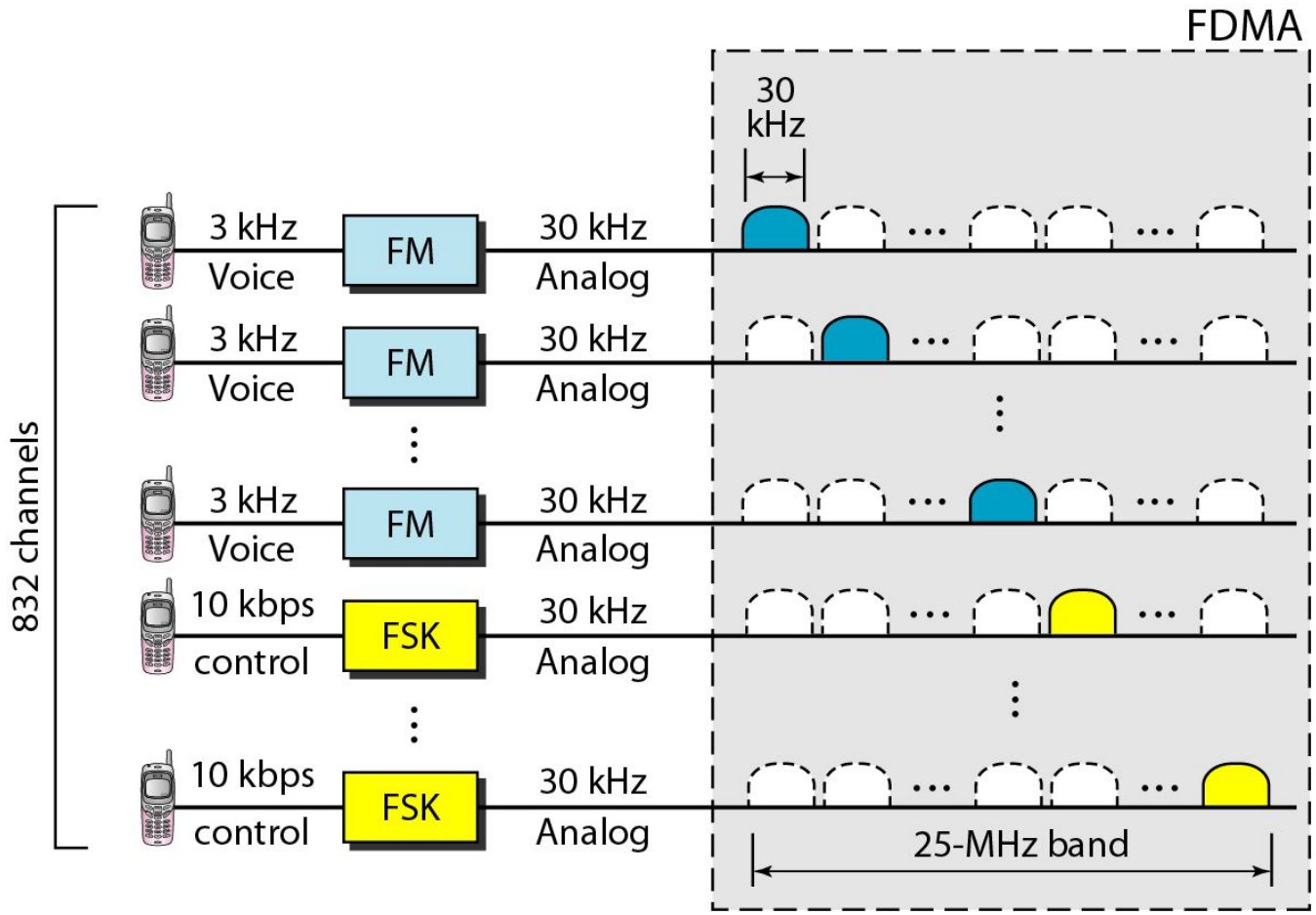


Figure 16.5 *Second-generation cellular phone systems*

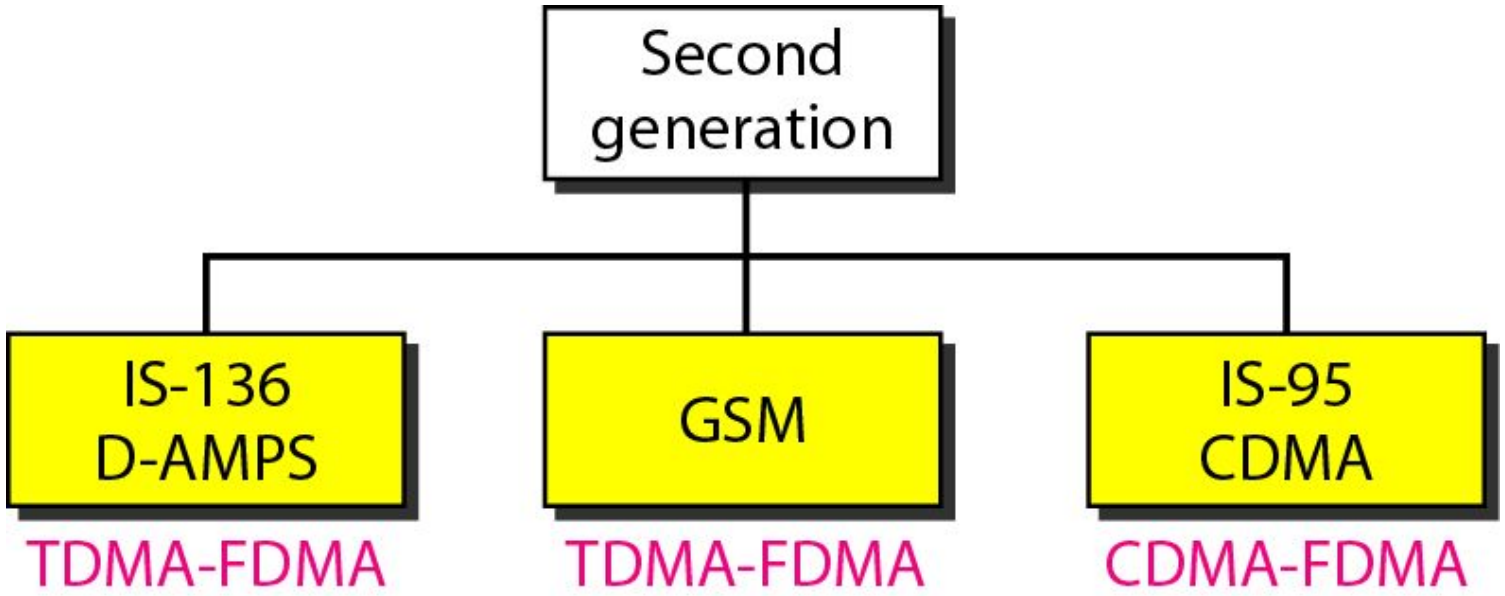
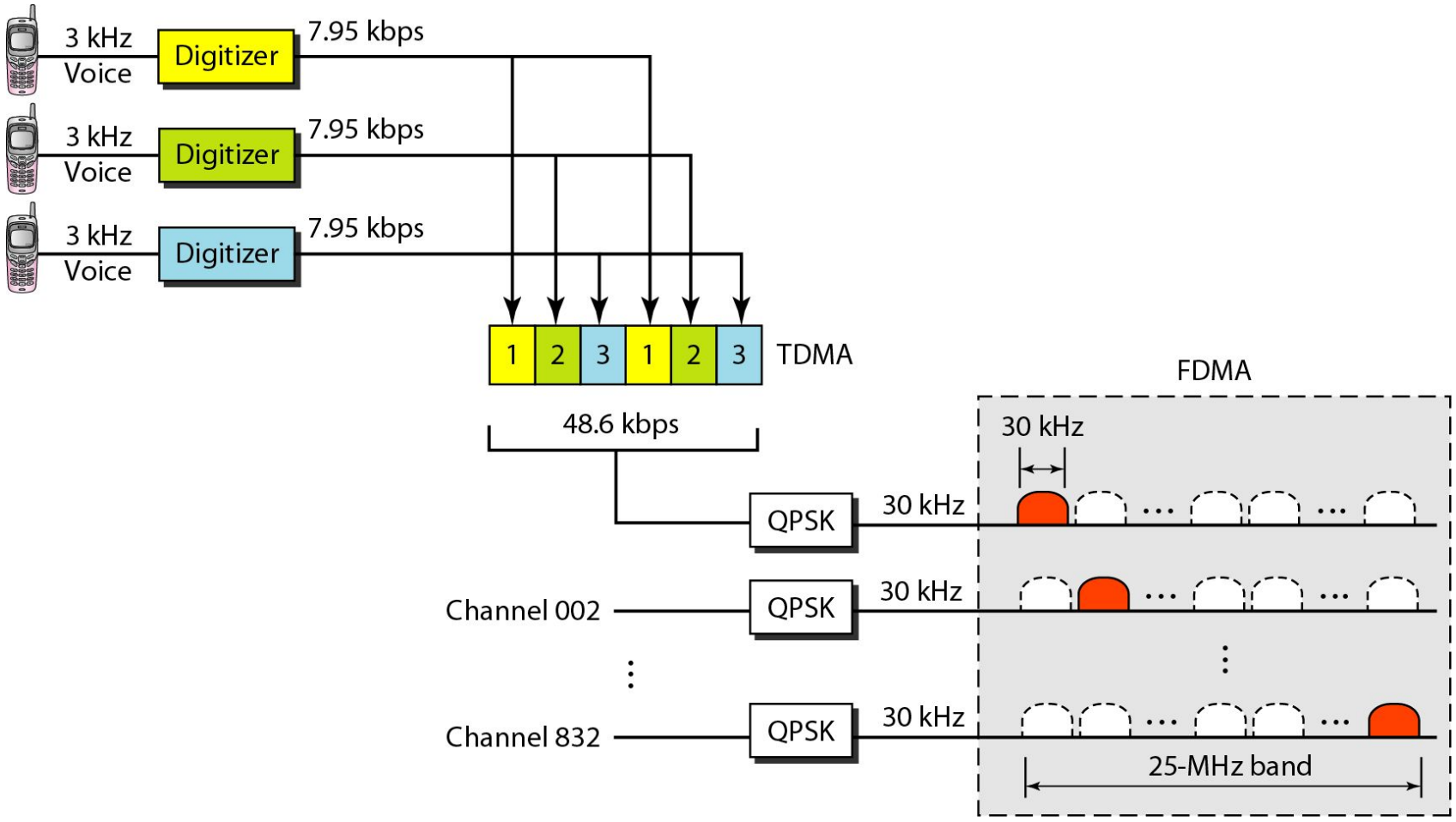


Figure 16.6 *D-AMPS*





Note

D-AMPS, or IS-136, is a digital cellular phone system using TDMA and FDMA.

Figure 16.7 *GSM bands*

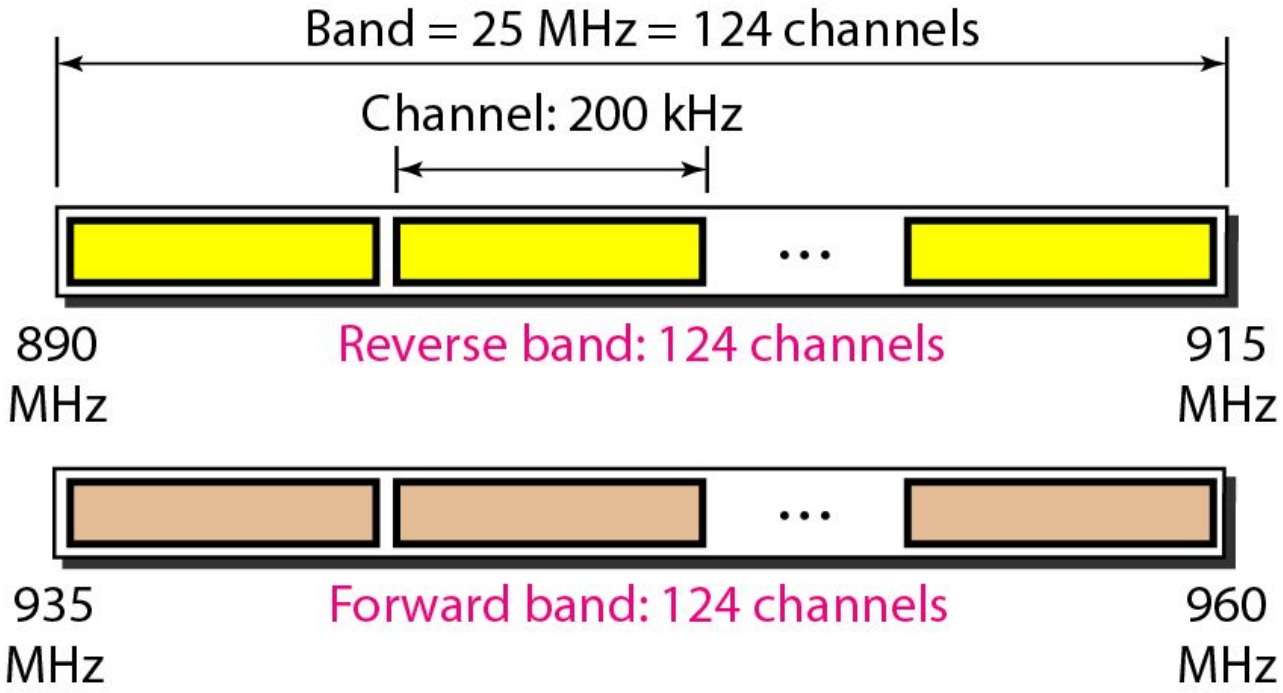


Figure 16.8 *GSM*

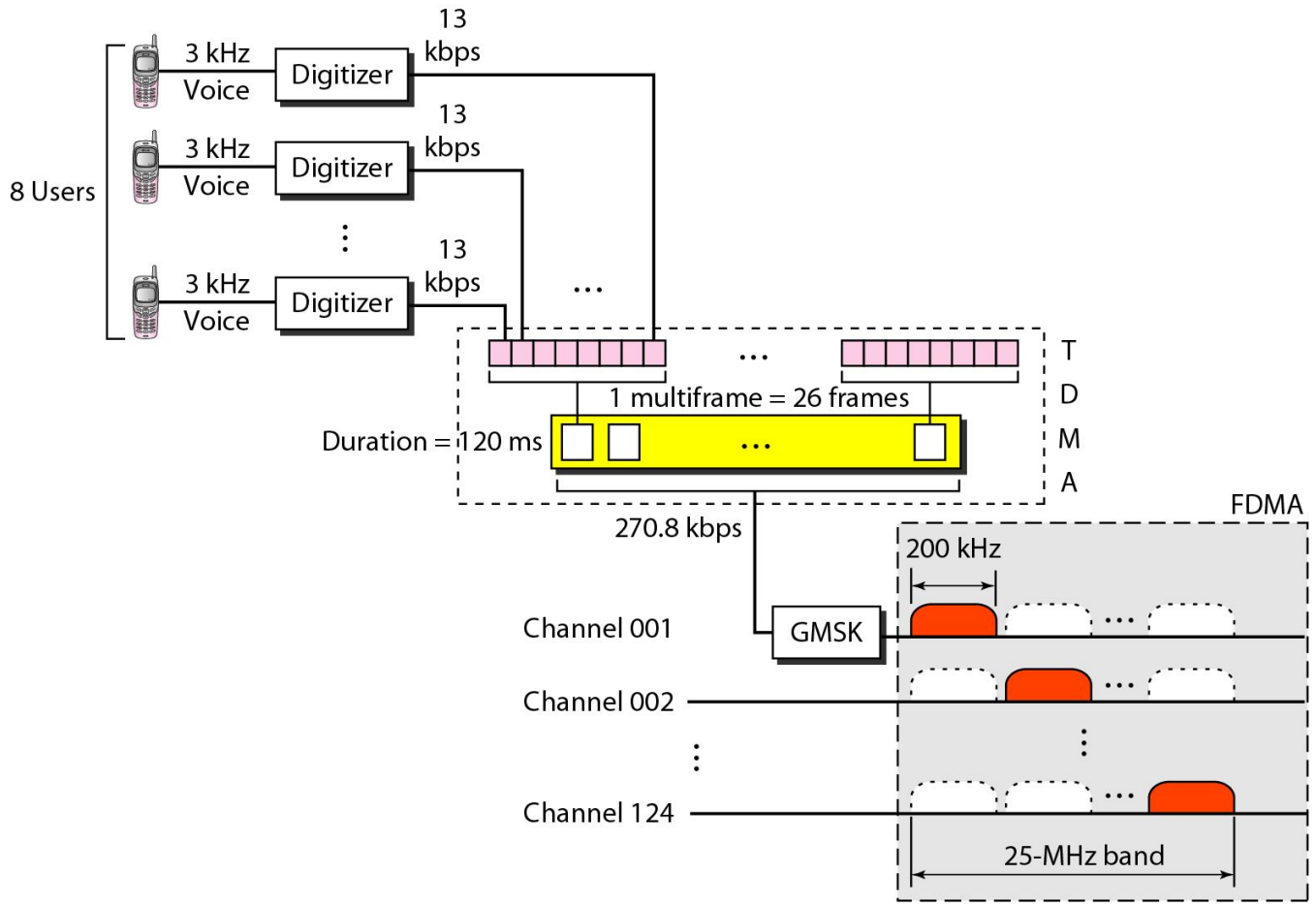
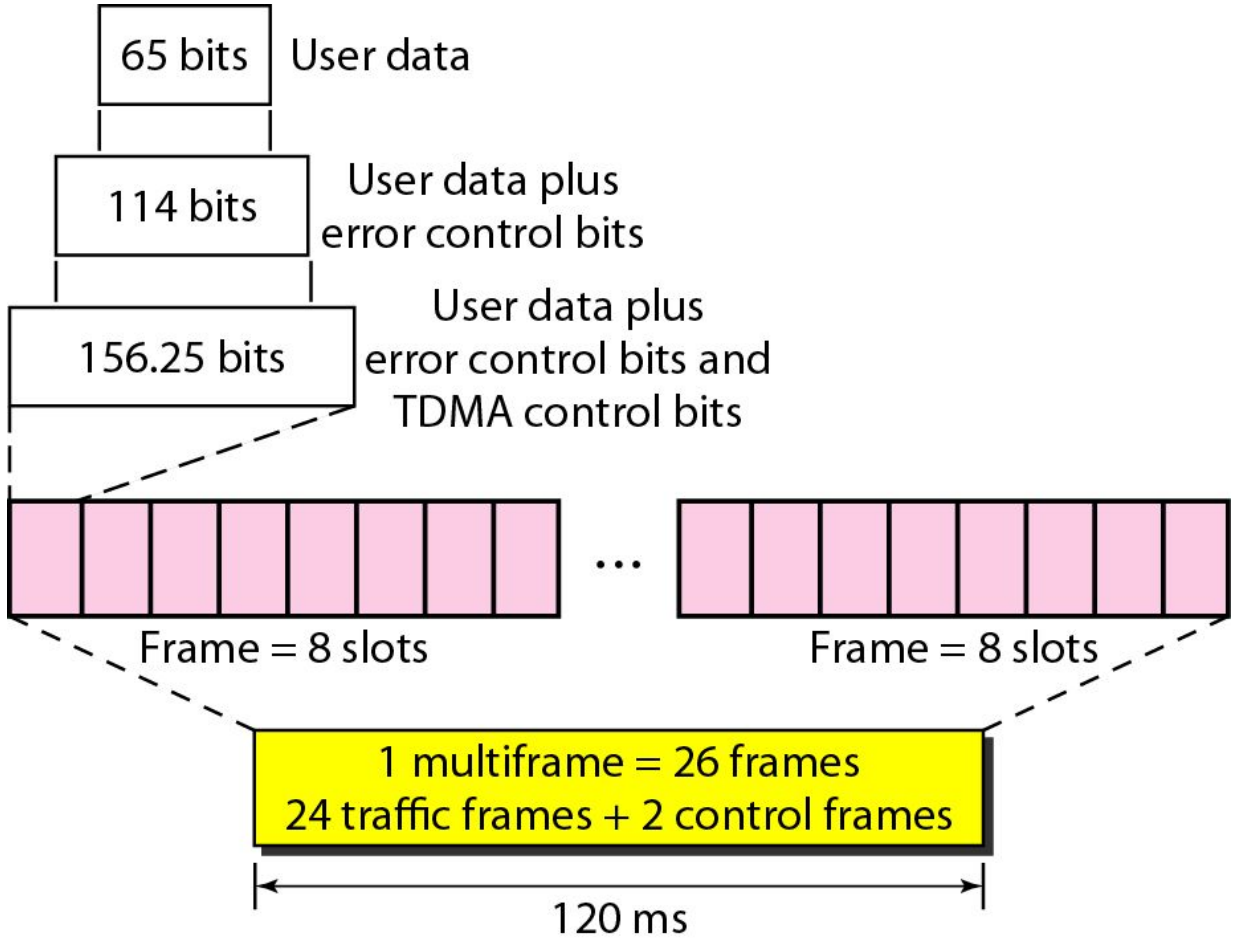


Figure 16.9 *Multiframe components*





Note

GSM is a digital cellular phone system using TDMA and FDMA.

Figure 16.10 *IS-95 forward transmission*

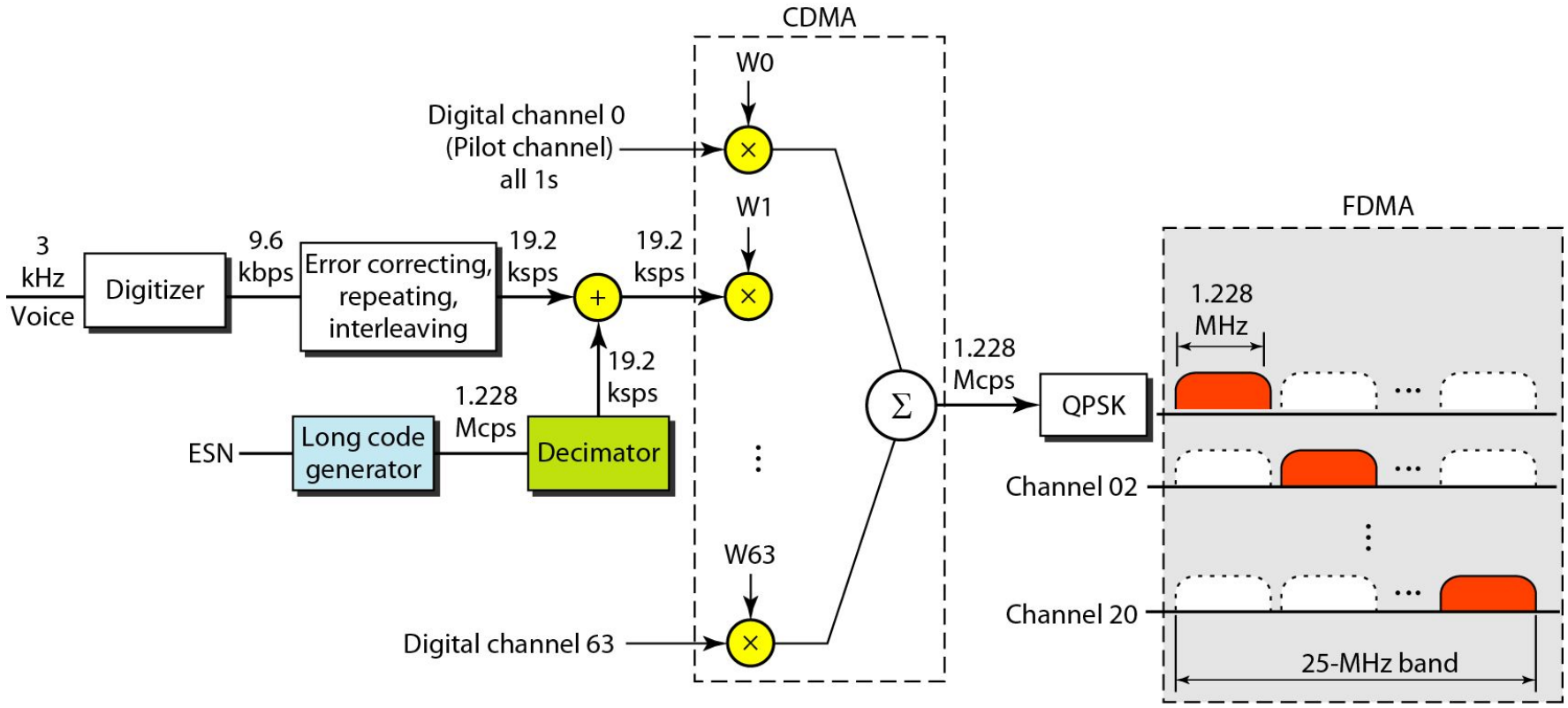
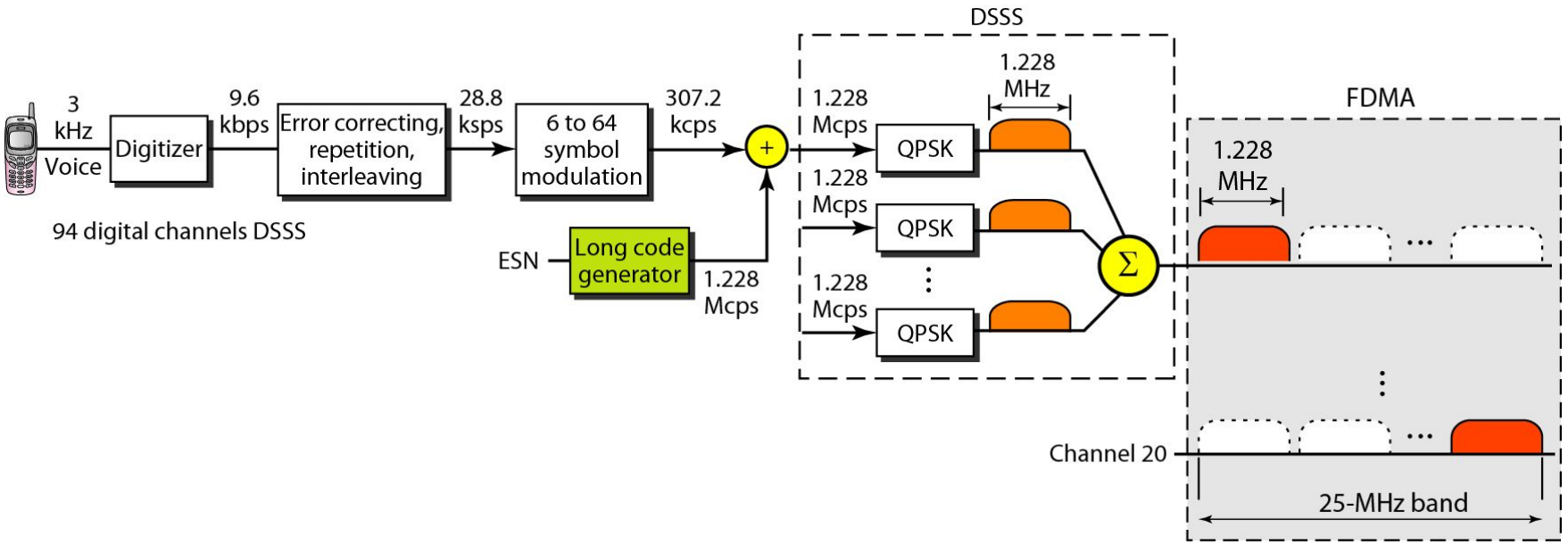


Figure 16.11 *IS-95 reverse transmission*





Note

IS-95 is a digital cellular phone system using CDMA/DSSS and FDMA.

Note

The main goal of third-generation cellular telephony is to provide universal personal communication.

Figure 16.12 *IMT-2000 radio interfaces*

